Lecture 7 – AI in Software Development

30 Exam-Style MCQs

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1. Artificial Intelligence (AI) is best defined as:

A. A programming paradigm for writing efficient code
B. Technology that enables machines to simulate human intelligence such as learning and decision making
C. A tool used only for automating testing
D. A subset of software engineering focused on optimization

Answer: B

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2. Generative AI differs from traditional AI mainly because it:

A. Executes predefined rules only
B. Focuses on numerical optimization problems
C. Can create original content such as text, code, or images
D. Works without any data

Answer: C

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3. Which of the following technologies is primarily used by Generative AI for text generation?

A. Expert systems
B. Decision trees
C. Large Language Models (LLMs)
D. Rule-based engines

Answer: C

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4. The worldwide surge in Generative AI adoption was mainly triggered by:

A. The invention of cloud computing
B. The release of GitHub Copilot
C. The arrival of ChatGPT in 2022
D. Advances in computer hardware

Answer: C

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5. In software development, AI-powered autocompletion mainly helps by:

A. Replacing developers completely
B. Predicting and generating the next lines of code
C. Designing system architecture automatically
D. Managing project budgets

Answer: B

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6. Boilerplate code refers to code that:

A. Is complex and performance-critical
B. Changes frequently between projects
C. Must be written repeatedly with little or no change
D. Is generated only during runtime

Answer: C

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7. Code synthesis using AI means that the AI:

A. Converts binary code into source code
B. Writes code based on natural language descriptions
C. Only reviews existing code
D. Executes code faster

Answer: B

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8. Which of the following is an example of AI-assisted bug detection?

A. Manual debugging using print statements
B. Predicting potential index out-of-range errors

C. Writing test cases by hand
D. Refactoring code manually

Answer: B

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9. Error prediction tools powered by AI primarily rely on:

A. Compiler warnings only
B. Historical code patterns and data
C. User interface testing
D. Hardware performance metrics

Answer: B

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10. One major advantage of AI-driven testing automation is that it:

A. Eliminates the need for requirements
B. Generates adaptive and comprehensive test cases
C. Removes the need for developers
D. Only tests user interfaces

Answer: B

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11. AI-based test optimization focuses on:

A. Running all tests every time
B. Removing failing tests
C. Prioritizing critical tests to save time and resources
D. Eliminating manual testing completely

Answer: C

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12. In documentation, Generative AI is mainly useful because it:

A. Writes code comments only
B. Automatically generates and updates readable documentation
C. Removes the need for APIs
D. Replaces technical writers entirely

Answer: B

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13. AI-assisted refactoring improves software mainly by:

A. Changing program behavior
B. Improving code readability, structure, and maintainability
C. Increasing code size
D. Adding new features

Answer: B

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14. Performance optimization using AI aims to:

A. Increase memory usage
B. Reduce functionality
C. Improve speed and resource efficiency without changing behavior
D. Replace algorithms

Answer: C

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15. Which security threat can AI tools help detect by analyzing suspicious input patterns?

A. Deadlock
B. SQL Injection
C. Infinite loops
D. Stack overflow

Answer: B

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16. AI-driven code auditing mainly ensures that code:

A. Compiles successfully
B. Is written in a specific programming language
C. Follows secure coding standards
D. Has detailed comments

Answer: C

17. In the Software Development Life Cycle (SDLC), AI can contribute to:

A. Only the coding phase
B. Only testing and debugging
C. Every phase from requirements to maintenance
D. Deployment only

Answer: C

18. During requirement gathering, Generative AI helps primarily by:

A. Writing final code
B. Extracting and clarifying requirements from natural language
C. Designing databases
D. Running automated tests

Answer: B

19. AI helps clarify ambiguous requirements by:

A. Ignoring vague statements
B. Removing non-functional requirements
C. Suggesting measurable and testable constraints
D. Automatically implementing them

Answer: C

20. In design and planning, AI tools can generate:

A. Executable binaries
B. UML diagrams and UI mockups from text descriptions
C. Only backend code
D. Deployment scripts only

Answer: B

21. Maintenance and support benefit from AI mainly because AI can:

A. Stop software updates
B. Predict failures and analyze logs continuously
C. Eliminate user feedback
D. Replace customer support teams fully

Answer: B

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22. Prompt Driven Development (PDD) is best described as:

A. Writing code without documentation
B. Using prompts as specifications for AI-generated code
C. Manual coding followed by testing
D. Traditional waterfall development

Answer: B

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23. In PDD, the role of the developer mainly shifts to:

A. Writing low-level assembly code
B. Crafting, reviewing, and refining prompts and outputs
C. Eliminating testing
D. Avoiding code reviews

Answer: B

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24. One major advantage of LLM-native IDEs over chat-based tools is that they:

A. Work without internet
B. Automatically deploy applications
C. Provide direct access to project code context and editing
D. Eliminate prompting

Answer: C

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25. The principle “Garbage in, garbage out” in prompting means that:

A. AI cannot make mistakes
B. Poor prompts lead to poor generated code
C. Prompts are optional
D. AI improves prompts automatically

Answer: B

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26. High-quality prompts should emphasize:

A. Speed, randomness, and flexibility
B. Context, specificity, and narrow scope
C. Broad scope and minimal detail
D. Ambiguity and creativity

Answer: B

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27. Prompt chaining is mainly used to:

A. Reduce AI usage
B. Solve complex tasks by breaking them into smaller steps
C. Increase hallucinations
D. Combine multiple programming languages

Answer: B

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28. One disadvantage of using a single large prompt for complex tasks is:

A. Faster execution
B. Better accuracy
C. Instruction neglect and error propagation
D. Reduced context

Answer: C

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29. The statement “LLMs write bugs just like humans” implies that:

A. AI-generated code should be merged without review
B. AI-generated code must be tested and reviewed carefully
C. AI always produces incorrect code
D. Testing is unnecessary

Answer: B

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30. The primary educational value of AI in software construction is that it:

A. Eliminates the need to learn programming
B. Encourages developers to focus on higher-level design and reasoning
C. Replaces software engineers
D. Removes the need for software processes

Answer: B